THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sum of the first 30 odd numbers is 900. The sum of the odd numbers from 21 to 59 is: 1

(a) 800 **(b)** 400 **(c)** 890 **(d)** 779

2. A spinner starts pointing North and turns clockwise 270° then anticlockwise 90°. The final angle clockwise from North is: 1

(a) 170 **(b)** 270 **(c)** 180 **(d)** 190

3. The number of 4-digit palindromes with first digit 2 is: 1

(a) 10 **(b)** 100 **(c)** 9 **(d)** 90

4. A bar graph has scale 1 cm = 12 units. The heights of two bars differ by 7 cm. The values differ by: 1

(a) 96 **(b)** 84 **(c)** 19 **(d)** 72

5. The smallest number divisible by each of 6, 8 and 12 is: 1

(a) 25 **(b)** 30 **(c)** 23 **(d)** 24

6. A rectangular field 24 m by 10 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 96 **(b)** 120 **(c)** 144 **(d)** 130

7. A drum is $\frac{3}{4}$ full. 20 litres are removed and it becomes $\frac{1}{4}$ full. The capacity (litres) is: 1

(a) 41 **(b)** 40 **(c)** 50 **(d)** 39

8. A circle of radius 3 cm is drawn. A square of side equal to the diameter is drawn around it. The area of the square (cm^2) is: 1

(a) 39 **(b)** 36 **(c)** 42 **(d)** 33

9. The number of letters among A, H, B, I, C with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 1 **(b)** 4 **(c)** 3 **(d)** 2

10. On a number line, the distance between -14 and 9 is: 1

(a) 24 **(b)** 22 **(c)** 23 **(d)** 26

11. The difference between the 20th and the 19th square numbers is: 1

(a) 20 **(b)** 41 **(c)** 37 **(d)** 39

12. The smaller angle between the hands of a clock at 1 o'clock is 30°. The smaller angle between the hands at 11 o'clock is: 1

(a) 150 **(b)** 30 **(c)** 330 **(d)** 60

13. The sum of 10 consecutive counting numbers starting from 12 is: 1

(a) 165 **(b)** 120 **(c)** 175 **(d)** 220

14. A school bus carries 30 children on trip 1, 45 on trip 2 and 60 on trip 3. A pictograph uses one symbol for 15 children. The number of symbols needed for all three trips is: 1

(a) 11 **(b)** 8 **(c)** 10 **(d)** 9

15. The greatest number that divides 48, 72 and 120 leaving no remainder is: 1

(a) 23 **(b)** 25 **(c)** 24 **(d)** 26

16. A picture 30 cm by 20 cm is mounted on a frame of width 3 cm all around. The area of the frame alone (cm^2) is: 1

(a) 333 **(b)** 348 **(c)** 336 **(d)** 339

17. A rod 6 m long is cut into pieces of $\frac{3}{4}$ m each. The number of pieces is: 1

(a) 9 **(b)** 8 **(c)** 10 **(d)** 7

18. Two circles of equal radius 4 cm touch externally. The distance from the outer end of one to the outer end of the other along the line of centres (cm): 1

(a) 12 **(b)** 16 **(c)** 24 **(d)** 20

19. The sum of the angles of rotation of a square and a regular hexagon (in degrees) is: 1

(a) 150 **(b)** 160 **(c)** 180 **(d)** 140

20. Two integers add to -5 and one of them is 12. The other is: 1

(a) -19 **(b)** -17 **(c)** -15 **(d)** -7

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. $\frac{3}{4}$ of a number is 18. Find the number and its $\frac{1}{3}$. 2

22. A circle has centre O and radius 10 cm. Points are at distances 6 cm, 10 cm, 13 cm and 9 cm from O. Find how many lie inside, on and outside the circle. 2

23. A regular polygon has rotational symmetry with the smallest angle of 30° and each side 4 cm. Find the number of sides and its perimeter. 2

24. Find the sum of the integers from -6 to 10 inclusive. 2

25. The sum of the first n odd numbers is 961. Find n and the 31st odd number. 2

26. A ray rotates 2 full turns and 90° more. Find the total angle turned and the number of right angles in it. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A designer wants a symmetrical logo with 5 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 5 petals are placed. 3

28. A submarine at -30 m dives 50 m, rises 40 m, then dives 25 m. Find its position after each move and the total vertical distance moved. 3

29. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 39. Then find the sum of the odd numbers from 41 to 49. 3

30. At 2:00 the minute hand starts turning clockwise and turns through 90°. How many minutes pass, how many right angles does it turn, and what time does the clock then show? 3

31. A row of 8 numbers starts 2, 3 and each next number is the sum of the previous two. Find the 8th number and the sum of all 8 numbers. 3

OR

In a pictograph one symbol = 12. A shop sold 3 symbols of pens, and the number of notebooks is 2 times the pens. Find the notebooks sold and total items sold. 3

32. Find the greatest number that divides 76 and 100 leaving remainder 4 in each case. 3

OR

A square of side 12 cm and a rectangle of length 14 cm and breadth 10 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. At the start of a game a player has 5 points. In four rounds the points change by -8, +6, -9, +12. Find the score after each round, the final score, and the lowest score reached. 5

34. Two sequences: A is $5, 9, 13, 17$, ... and B is $20, 22, 24, 26$, ... Find the 12th term of each and decide which is larger and by how much. 5

35. A robot starts facing North. It turns clockwise 180°, then 45° anticlockwise, then 90° clockwise. Find its final direction angle clockwise from North and the further clockwise turn needed to face North again. 5

36. Find the number of steps the reverse-and-add process takes from 67 to reach a palindrome, write each sum, and state whether the final palindrome has an even or odd number of digits. 5

OR

100 students were asked their favourite colour: red 20, blue 25, green 15, yellow 10 and the rest chose pink. Find the number liking pink, the fraction liking blue, and draw the pictograph with one symbol = 5 (give symbol counts). 5

37. Find the smallest 4-digit number divisible by 6, 8 and 12. Also find the largest 4-digit number divisible by all three. 5

OR

A rectangular park 60 m by 40 m has two paths each 3 m wide crossing in the middle, one parallel to each side. Find the area of the paths and the area of the rest of the park. 5

38. Add $1\frac{1}{2}$, $2\frac{1}{3}$ and $3\frac{1}{6}$ and subtract $\frac{3}{4}$ from the sum. 5

OR

A designer draws 5 identical circles of radius 3 cm in a row, each touching the next. Find the total length of the row and the distance between the centres of the first and the last circle. How many circles of this size fit exactly in a strip 42 cm long? 5

END OF QUESTION PAPER — Total: 80 Marks